Name: Class: Date:

Activity 1 · Hands-on · 🧪 In the lab · ⏱ 2–3 hours

🌼 Grow Pollen Tubes

You will learn: On a sugary stigma, a pollen grain germinates and grows a pollen tube through its germ pore.

YOU NEED

- Fresh flowers: Vinca (sadabahar), balsam, hibiscus or pea
- Sugar and water (10% solution: 1 spoon sugar in 10 spoons water)
- A pinch of boric acid (optional, helps germination)
- Glass slide or clear sheet, a dropper
- A box lined with wet tissue
- Phone microscope

⚠️ **Safety:** Boric acid must not be eaten; wash hands after use.

STEPS

1. Put a drop of sugar solution on the slide.
2. Tap a ripe anther over the drop so pollen falls on it.
3. Keep the slide in the wet-tissue box (to stop drying) for 30 minutes.
4. Observe at high zoom; count how many of 20 pollen grains show a tube.
5. Check again at 1 and 2 hours.
6. Sketch a germinating pollen grain with its tube.

OBSERVATION TABLE

Time	Grains with a tube (out of 20)	% germination	Tube length (short / long)

QUESTIONS

1. Through which part of the exine does the pollen tube come out?

2. Why did we use sugar solution?

3. What does the pollen tube carry in the real flower?

WHAT IS HAPPENING? — THE SCIENCE

The sugar solution imitates the stigma. The pollen grain absorbs it and the vegetative cell grows out through a germ pore as the pollen tube — the same tube that carries the two male gametes to the embryo sac.

Key terms: pollen tube · germ pore · stigma · male gametes

Activity 2 · Hands-on · 🌳 Outdoors · ⏱ 45 minutes

🐝 Who Pollinates My Garden?

> **You will learn:** Flowers show adaptations for their pollinating agent — wind, insects or birds.

YOU NEED

- 5 different flowers in a garden (include grass or maize if possible)
- Notebook
- Phone camera
- Watch or timer

> ⚠️ **Safety:** Keep a safe distance from bees and wasps; do not disturb hives.

STEPS

1. Choose 5 flowers. For each, note colour, smell, size and nectar (touch a petal base).
2. Tap the flower and see if pollen is sticky or powdery.
3. Watch each flower for 5 minutes and count visitors (bees, butterflies, birds, ants).
4. Look at grass or maize flowers — notice the hanging anthers and feathery stigma.
5. Decide the pollinating agent for each flower.

OBSERVATION TABLE

Flower	Colour / smell	Nectar?	Pollen sticky or dry	Visitors in 5 min	Pollinated by

QUESTIONS

1. List three features of wind-pollinated flowers.

2. Why do insect-pollinated flowers produce nectar?

3. What is the advantage of cross-pollination?

WHAT IS HAPPENING? — THE SCIENCE

Insect-pollinated flowers are large, colourful, scented and have nectar and sticky pollen. Wind-pollinated flowers (grasses, maize) are small, dull, without nectar, with light, dry pollen, exposed anthers and feathery stigmas to catch pollen.

Key terms: **pollination** **anemophily** **entomophily** **nectar**

Digital activity

💻 From Pollen to Seed

> Tap the events in the correct order.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch1/2**

TAP THE STEPS IN THE CORRECT ORDER.

1. _____ Pollen tube grows down the style
2. _____ Pollen lands on the stigma (pollination)
3. _____ Syngamy: male gamete + egg → zygote (2n)
4. _____ Tube enters a synergid via the filiform apparatus
5. _____ Pollen grain germinates on the stigma
6. _____ Endosperm develops from PEN
7. _____ Triple fusion: male gamete + 2 polar nuclei → PEN (3n)
8. _____ Tube enters the ovule through the micropyle
9. _____ Two male gametes are released
10. _____ Ovule becomes seed; ovary becomes fruit
11. _____ Zygote develops into the embryo

> **You will learn:** Two fusions in one embryo sac — syngamy and triple fusion — make double fertilisation, unique to flowering plants.

THINK & WRITE

1. What is the ploidy of the endosperm? Why?

2. Which cells guide the pollen tube into the embryo sac?